

```
1 import java.util.*;
2 class CartonBoxes
3 {
4     int n;
5     void stockcheck(int n)
6     {
7         if ((n < 1) || (n > 1000))
8         {
9             System.out.println("INVALID INPUT");
10            System.exit(0);
11        }
12    }
13
14    void compute(int n)
15    {
16        stockcheck(n);
17        int cartonSizes[] = {48, 24, 12, 6};
18        int total = 0;
19        int t = n;
20        System.out.println(t);
21        for (int i = 0; i < cartonSizes.length; i++)
22        {
23            int cartonCount = t / cartonSizes[i];
24            t = t % cartonSizes[i];
25            total += cartonCount;
26            if (cartonCount != 0)
27            {
28                System.out.println(cartonSizes[i] + " * " + cartonCount +
29                    " = " + (cartonSizes[i] * cartonCount));
30            }
31        }
32
33        /*
34         * This if check is for the case when
35         * boxes left are less than 6. We need
36         * one more carton of capacity 6 in this
37         * case so total is incremented by 1.
38         */
39        if (t != 0) {
40            System.out.println("Remaining boxes = " + t
41                + " * 1 = " + t);
42            total++;
43        }
44        else
45        {
46            System.out.println("Remaining boxes = 0");
47        }
48        System.out.println("Total number of boxes = " + n);
49        System.out.println("Total number of cartons = " + total);
50    }
```

51 |
52 | }